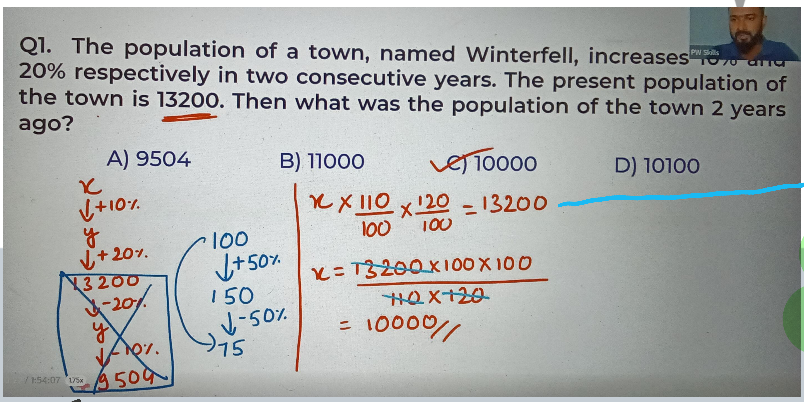

Imp.

either use this or agar kuch na samjhe to options ko put karke nikaal lo

Do NOT make a mistake of making equations jo meri aadat hai time waste bhi hoga or banega bhi nahi.

Quantitative Aptitude (simple interest, compound interest and percentages)

(1.) A% of B = B% of A

ex → 16% of 50 = 50% of 16

$\frac{16}{100} \times 50 = \frac{50}{100} \times 16$

8 = 8

(2.) A% of X ± B% of X

= (A ± B)% of X

ex → 10% + 20% is invalid. % don't mean anything on it's own. So,

10% of 500 + 20% of 500 = 30% of 500

(3.) Increase by A% = (100 + A)%

ex → +10% ⟹ 110%

$500 + 10\% = \frac{110}{100}$ of 500 = 550.

Decrease by B% = (100 − B)%

ex → −20% ⟹ 80%

$500 - 20\% = \frac{80}{100}$ of 500

= 400.

$$\text{Change\%} = \frac{|\text{Change}|}{\text{Old value}} \times 100$$

(∵ Change = New value − Old value)

Q2. Salary of Monica is 25% more than Rachel's. Then the salary of Rachel is what percentage less than Monica's?

A) 25% B) 20% C) 16.66% D) 33.33%

Let R = 100

M = 125

$\frac{25}{125} \times 100 = 20\%$

Q3. When a number is first increased by 10% and then reduced by 10%, the number:

A) Does not change
B) Decreases by 1%
C) Increases by 1%
D) None of these

100

↓ +10%

110 −1%

↓ −10% = 11

99

Q7. If X and Y are 20% and 25% greater than Z respectively, by how much percentage is

i. X smaller than Y?

A) 20% B) 4% C) 5% D) 4.16%

ii. Y greater than X?

A) 20% B) 4% C) 5% D) 4.16%

Z = 100

X = 120

Y = 125

$\frac{5}{125} \times 100$ = 4%

$\frac{5}{120} \times 100 = 4.16\%$.

Bada ya chota hai upar wo number, niche than ke baad wala number × 100

Imp

Q1. Hulk mistakenly divided a number by 2 instead of multiplying it by 2. Find the percentage of error.

A) 35% B) 45% C) 65% D) 75%

N = 100

$W = \frac{100}{2} = 50$

$C = 100 \times 2 = 200$

$\% E = \frac{150}{200} \times 100 = 75\%$

(Correct value in the denominator)

Q5. If the price of petrol increases by 25%, by how much must Batman cut down his consumption so that his expenditure on petrol remains constant?

A) 25% B) 16.67% C) 20% D) 33.33%

Let $P = 100$, $P_2 = 125$

$C = 100$, $C_2 = ?\ (80)$ — 20%

$E = 100 \times 100 = 10000$

$E_2 = 10000$

$P \times C = E$

$C_2 = \frac{E_2}{P_2} = \frac{10000}{125} = 80$

(same like Q2)

same

Q2. If the price of petrol increases by 50% and Stark intends to spend only an additional 25% on petrol, by how much will he reduce the quantity of petrol purchased?

A) 25% B) 16.66% C) 50% D) 20%

Let $P = 100$, $P_2 = 150$

$C = 100$, $C_2 = ?$ — 16.66

$Exp = 100 \times 100 = 10000$

$Exp_2 = \frac{125}{100} \times 10000 = 12500$

$C_2 = \frac{12500}{150} = 83.33$

$P = 100$, $P_2 = 150$

$E = 100$, $E_2 = 125$

$\frac{25}{150} \times 100 = 16.66\%$

Q3. In an election between two candidates, 20% of votes were declared invalid. First candidate got 70% of the valid votes and a lead of 1600 votes. The total number of votes enrolled in that election was:

A) 5000 votes B) 5400 votes C) 10000 votes D) 6667 votes

T = 100

↓ −20% = INVALID

80 VALID

W 70% — 30% L

56 − 24 = 32 = Lead

Lead	Total
32	100

100 votes me 32 ki lead.

to,

$\Rightarrow \frac{100}{32}$ (1 ki lead me),

$\Rightarrow \frac{100}{32} \times 1600$ (1600 ki lead me)

$\Rightarrow$ 5000.

Q4. Fresh grapes contain 90% water by weight while dried grapes contain 20% water by weight. What is the weight of dry grapes available from 20 kg of fresh grapes?

A) 2 kg B) 2.4 kg C) 2.5 kg D) None of these

FRESH: 90% W, 10% P, 20 kg — Pulp 2 kg — DRY: 20% W, 80% P, x kg

Pulp Fresh = Pulp Dry

10% of 20 kg = 80% of x kg

$2 = \frac{80}{100} \times x$ | $x = \frac{2 \times 100}{80} = 2.5$ kg